



SRM INSTITUTE OF SCIENCE AND TECHNOLOGY,
Ramapuram, Chennai-89
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



21CSC203P - APP
First Review

II year /III Sem

Class/Sec : B.Tech- Information Technology

**TREKMATE- TREKKING MANAGEMENT
APPLICATION**

Batch No.:

Team Members	Supervisor
Ananth R - RA2511008020022 Dhillip Prasanna - RA2511008020001 Shyam Vignesh - RA2511008020019 Amish W.S - RA2511008020034	NAME:

Date

Slide Number

AGENDA

- Introduction
- Abstract
- Scope and Motivation
- Objectives
- Literature Review (Chronological Order)
- Architectural Diagram
- ER diagram
- Database Connectivity
- List of Modules (detailed Manner)
- References (Base Paper to be included)

Introduction

The Trekking Management Application is a centralized, web-based platform designed to simplify and digitize the planning, booking, management, and monitoring of trekking activities.

It connects trekkers, guides / organizers, and administrators through a single platform — offering trek discovery and booking, participant management, live location tracking, weather alerts, SOS / emergency assistance, reviews, analytics, and AI-powered query answering.

The application aims to make trekking safer, more organized, accessible, and efficient for every stakeholder involved.

1

Trekker

Discovers, books & tracks treks

2

Guide / Organizer

Manages treks & participants

3

Admin

Oversees users, data & reports

Abstract

Traditional trekking management relies on multiple disconnected platforms for discovery, booking, communication, navigation, weather and emergency support — resulting in poor coordination, limited real-time visibility, and difficulty managing participants and emergencies.

The proposed system integrates the following into a single platform:

- Trek discovery, booking & preparation info
- Participant management & trek-progress tracking
- JWT authentication & PostgreSQL data management
- Route monitoring, weather updates & communication
- GPS-based live tracking & WebSocket real-time updates
- AI-powered RAG + NLM query-answering assistant

Designed to improve safety, coordination, and the overall trekking experience for all stakeholders.

Scope and Motivation

MOTIVATION

- Difficulty finding reliable trek information
- Manual or fragmented booking processes
- Lack of real-time participant tracking
- Limited communication between trekkers & guides
- Unexpected weather conditions
- Difficulty handling emergency situations
- Lack of centralized trek management
- Difficulty accessing relevant information quickly

SCOPE

Trek Discovery

Browse treks, view details, difficulty & availability

Online Booking

Booking, participants, payment & confirmation

Trek Management

Organizer dashboard, create & monitor treks

Real-Time Safety

GPS tracking, SOS, weather alerts, live chat

AI Assistance

NL queries, RAG retrieval, recommendations

Administration

User & guide management, reviews, analytics

Objectives

PRIMARY OBJECTIVE

To develop an integrated trekking management platform that improves trek planning, booking, coordination, safety, and monitoring.

- 1

Centralized platform for discovering and booking treks
- 2

Simplified trek & participant management for organizers
- 3

Real-time GPS-based participant tracking
- 4

Weather information & safety alerts
- 5

SOS / emergency alert mechanism
- 6

Real-time communication between trekkers & organizers
- 7

AI-based answers to trekking-related queries
- 8

Secure authentication & role-based authorization
- 9

Monitoring, reporting & analytics for administrators
- 10

Improved overall safety and convenience of trekking

Literature Review

LITERATURE REVIEW (SHORTENED – 4 STUDIES)

YEAR	TITLE	KEY CONTRIBUTION	LIMITATION	RELEVANCE TO OUR SYSTEM
2018 	Survey on Location-Based Services and Applications <i>International Journal of Computer Applications, Vol. 179, No. 45, 2018</i>	 Reviewed GPS and location-based services used in tourism and travel.	 Did not address real-time group tracking or emergency management.	 Supports GPS-based live tracking and map integration.
2020 	IoT-Based Health Monitoring and Safety System <i>IEEE Access, Vol. 8, pp. 227056–227067, 2020</i>	 Proposed a wearable system for monitoring health parameters and sending safety alerts.	 Requires additional hardware and lacks trek management features.	 Provides ideas for future health and safety monitoring features.
2023 	Large Language Models (LLMs) <i>arXiv: 2303.08774, 2023</i>	 Demonstrated how LLMs can understand and generate human-like responses.	 May provide outdated information and lacks domain-specific knowledge.	 Forms the foundation of the AI-powered query system.
2025 	Integrated Trekking Management Systems <i>Journal of Outdoor Recreation and Tourism, Vol. 45, 2025</i>	 Identified the need for a platform that combines tracking, booking, safety, and AI.	 Existing systems do not integrate all features into a single platform.	 Validates the research gap and the need for an integrated trekking management system.



RESEARCH GAP IDENTIFIED: Existing trekking applications focus on only one or two features. There is a need for a **single integrated platform** that combines:



Trek Management



Booking



Real-Time Tracking



Safety & SOS Alerts



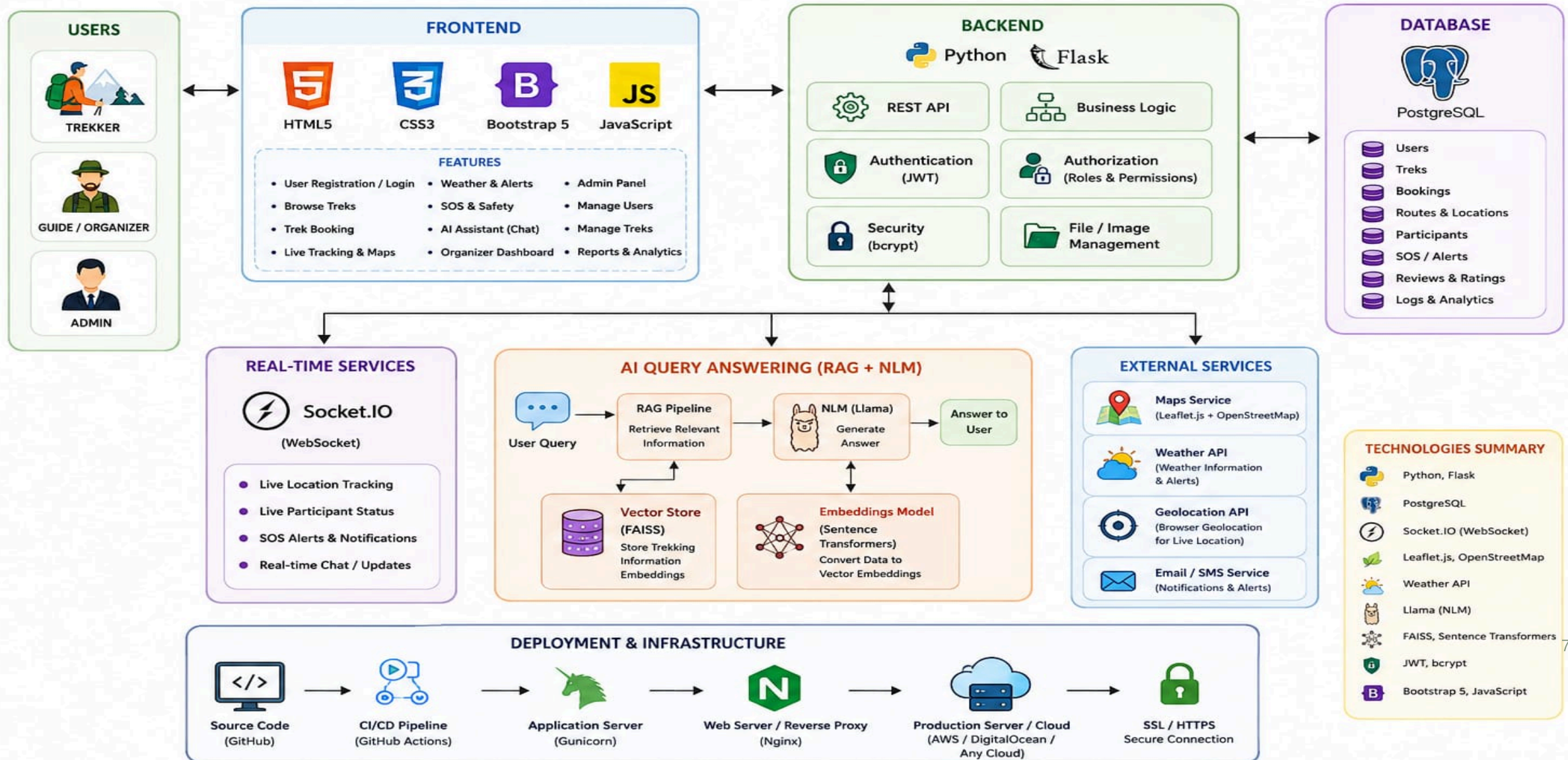
Weather Updates



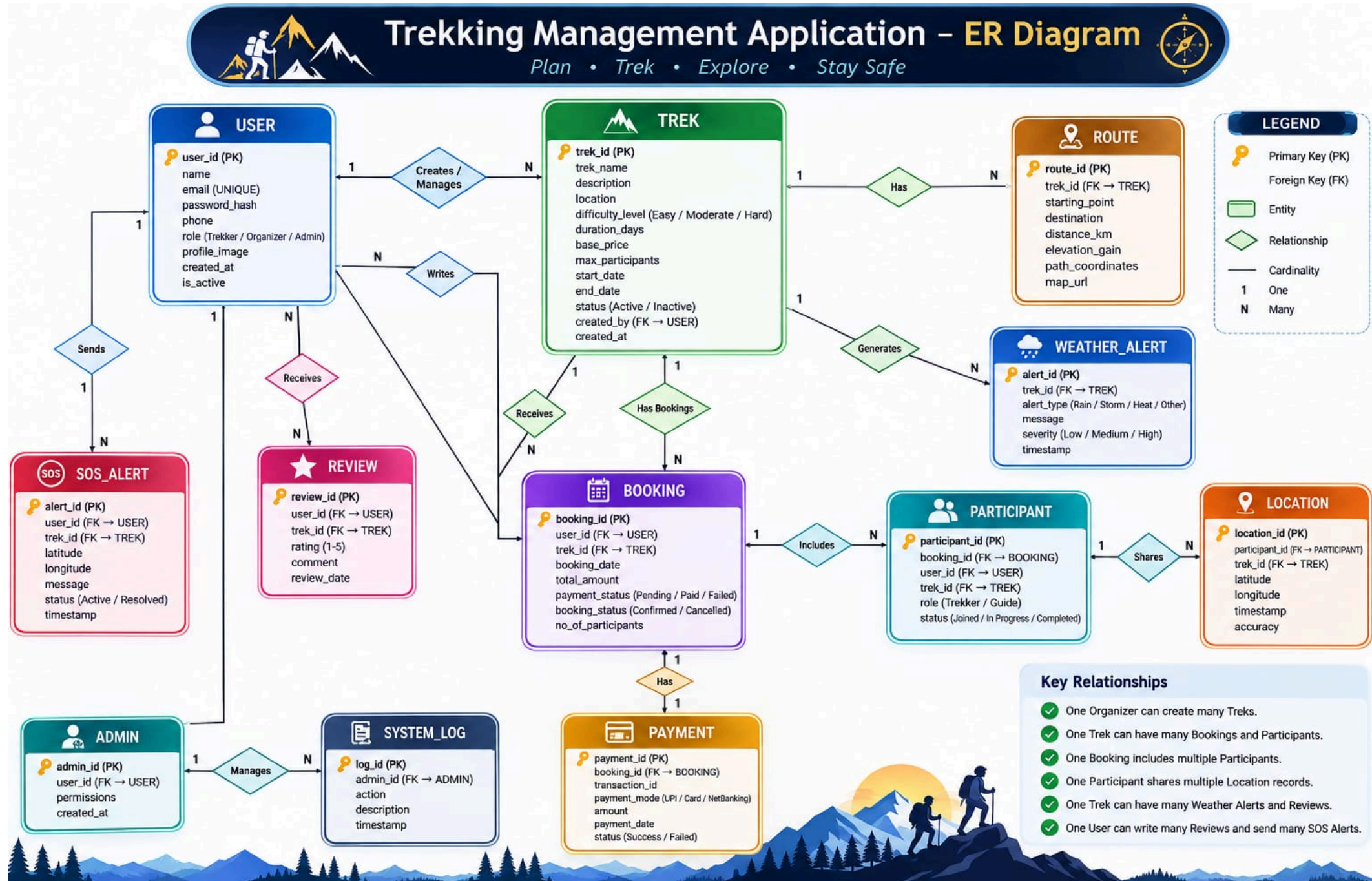
AI-Based Query Answering

System Architecture Diagram

TREKKING MANAGEMENT APPLICATION SYSTEM ARCHITECTURE

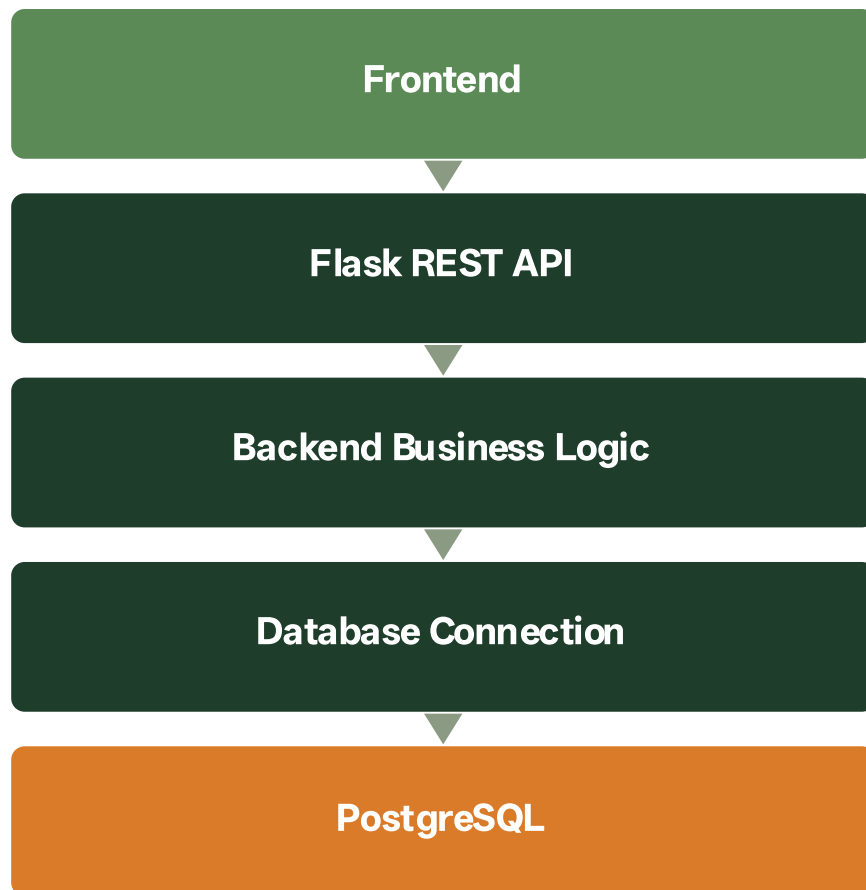


ER Diagram



Database Connectivity

PostgreSQL is used as the primary relational database, accessed through a layered Flask REST API architecture.



DATABASE OPERATIONS

INSERT

New users, bookings, reviews, and alerts

SELECT

Trek information, users, and bookings

UPDATE

Booking status, trek info, participant status

DELETE

Removing or managing records

Key tables: Users · Treks · Bookings · Routes & Locations · Participants · SOS / Alerts · Reviews & Ratings · Logs & Analytics

List of Modules

01

User Management

Registration, login, roles (Trekker / Guide / Admin), JWT authentication & authorization

02

Trek Discovery & Booking

Browse treks, view details, book, manage participants & payments

03

Trek & Organizer Dashboard

Create, manage and monitor treks and participant progress

04

Real-Time Tracking & Safety

GPS live tracking, SOS alerts, weather alerts & notifications

05

Real-Time Communication

WebSocket-based chat, live status & updates via Socket.IO

06

AI Query Assistant

RAG + NLM powered natural-language trekking Q&A

07

Reviews & Ratings

Post-trek feedback, ratings & trek reputation

08

Admin & Analytics

User/trek management, reports, logs & system analytics

References

[1]

G. Sharma, R. K. Bedi, "A Survey on Location Based Services and Applications," International Journal of Computer Applications, Vol. 179, No. 45, 2018.

[2]

A. Kumar, P. Kumar, "Mobile Based Emergency Alert System," International Research Journal of Engineering and Technology (IRJET), Vol. 06, Issue 03, 2019.

[3]

M. H. Islam, M. R. Islam, "IoT Based Health Monitoring and Safety System for Adventurers," IEEE Access, Vol. 8, pp. 227056–227067, 2020.

[4]

S. Verma, R. Jain, "Recommender Systems in Tourism: A Review," Journal of Tourism & Hospitality, Vol. 10, Issue 2, 2021.

[5]

J. Li, H. Zhang, "Real-Time Web Applications Using WebSocket," IEEE Internet Computing, Vol. 26, Issue 1, pp. 72–79, 2022.

[6]

T. Brown et al., "Language Models are Few-Shot Learners" (GPT-4 Technical Report), arXiv:2303.08774, 2023.

[7]

P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS 2024, Vol. 37, 2024.

[8]

K. Singh, P. Thomas, "Integrated Trekking Management Systems: A Systematic Review," Journal of Outdoor Recreation and Tourism, Vol. 45, 2025.